

Booking.com's Agentic Ecosystem

A builder's perspective on the journey

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THE JOURNEY BEGINS

Let's rewind to 2023.

THE PIVOTAL MOMENT

Conversation became **an API.**

THE QUESTION

**We started asking,
what could it mean for travel?**

THE INDUSTRY RESPONSE

Two industry patterns

Extend ChatGPT with Plugins

Embed ChatGPT in a website

OUR PRODUCT AMBITION

A deeper experience beyond chat

Travel-specific, personalized, and grounded in Booking.com content.

THE FIRST USE CASE

We picked **Discovery**.

A low-regret surface with high strategic upside and customer value.

THE STATUS QUO

Travellers translated **intent** into **search filters.**

Destination · Dates · Filters

THE PRODUCT OPPORTUNITY

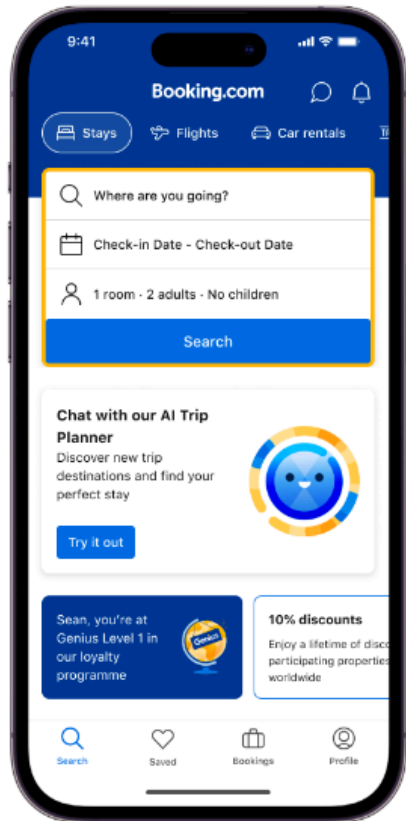
**But ... what if travellers could
describe their trip?**

THE FIRST PRODUCT

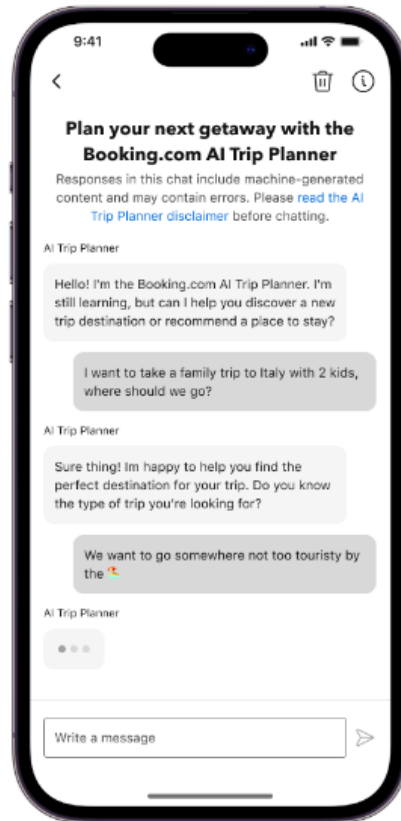
Booking.com's AI Trip Planner

AI Trip Planner

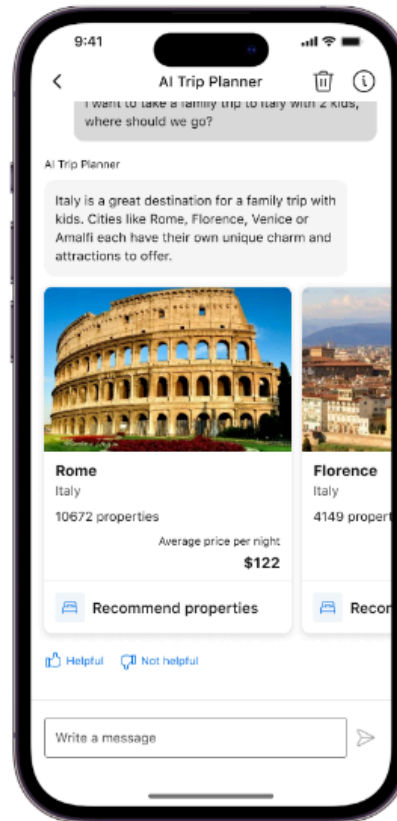
Entry Point



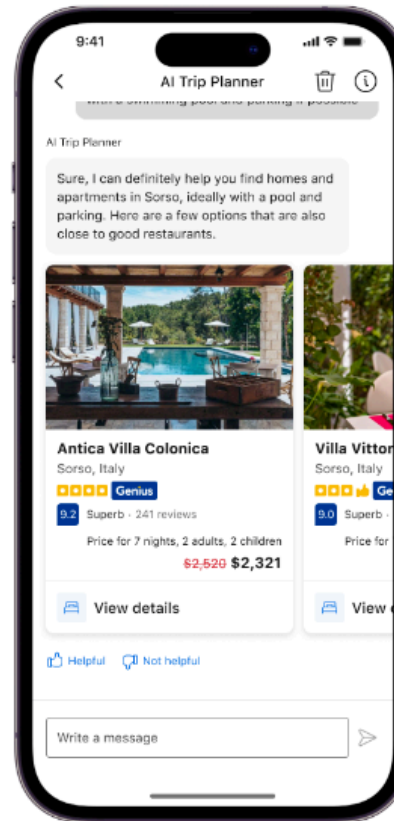
Conversational Interface



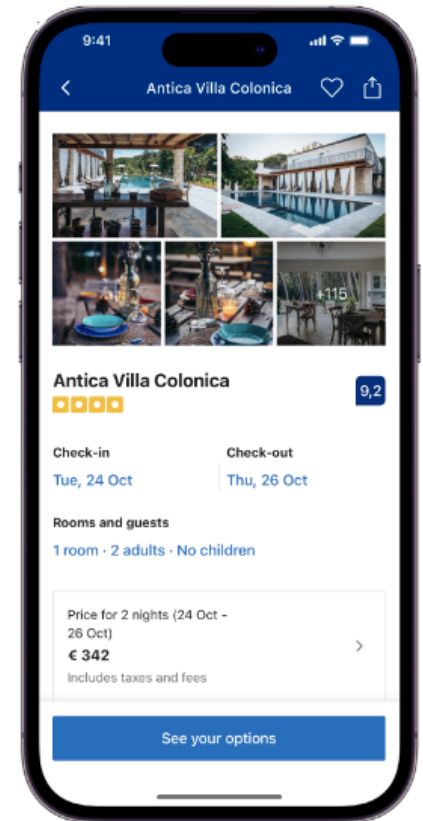
Destination Recommendation



Property Recommendation



Property Page



THE PRODUCT GOAL

**Not to chat but to
help travellers move forward**

Expand the top of the funnel

THE FOUNDATION

We were not starting from **zero**.

~20 years of ML expertise

BEFORE THE GENAI WAVE

We had built **AI chatbots** before.

Booking Assistant · 2017

THE FOUNDATION

And, we had mature ML Platforms



Recommendation in action

Dreaming of your next trip?
Add some top-rated spots to your itinerary

Alkmaar

Paris

Amsterdam

Recently viewed

Los Angeles Athletic Club...
West Hollywood, Los Angeles
4.9 Fabulous - 1,234 reviews
6 nights €1,321 €1,294

Amsterdam → Lisbon
09:15 AMS - 25 Sep
12:45 LIS - 25 Sep
Transavia Airlines
Direct • 2h 30m

15:20 LIS - 26 Sep
18:50 AMS - 26 Sep
Transavia Airlines, Lufthansa
2 stops • 3h 30m
Round trip €350

Warner Bros. Studio Tour London
Hollywood, Los Angeles
4.9 Exceptional - 32 reviews
Adult tickets from €45

Stay at our top unique properties
From castles and villas to boats and igloos, we have it all

Casaprisco
Netherlands, Putten
4.9 Wonderful - 51 reviews

Regina Vught
Netherlands, Vught
4.8 Very Good

B&B Puur Drenthe
Netherlands, Elp
4.9 Excellent

Luxe Eco Pod op Stal
Netherlands, Drijber

Browse by specific property types
Explore unique property types in Paris...

Boutique Hotels
650 available properties

Artist Apartments
800 available properties

Houseboats
300 available properties

Rooftop
1,650 available properties

Things to do in and around London **Genius** Up to 20% off
Save up to 20% on select attractions [See all](#)

Admission to the London Eye
★ 4.5 (5528 reviews)
Tickets from €28.22

Admission to Madame Tussauds London
★ 4.6 (3582 reviews)
Tickets from €21 €28.15

Tower of London and Crown Jewels Tour
★ 4.43 (3746 reviews)
Tickets from €42.86

Popular flights near you
Find deals on domestic and international flights

Amsterdam to London
12 Mar - 19 Mar - Round trip

Amsterdam to Milan
12 Mar - 19 Mar - Round trip

Amsterdam to Barcelona
14 Mar - 21 Mar - Round trip

Amsterdam to Copenhagen
12 Mar - 19 Mar - Round trip

THE 2023 REALITY

Building at the frontier came with real constraints.

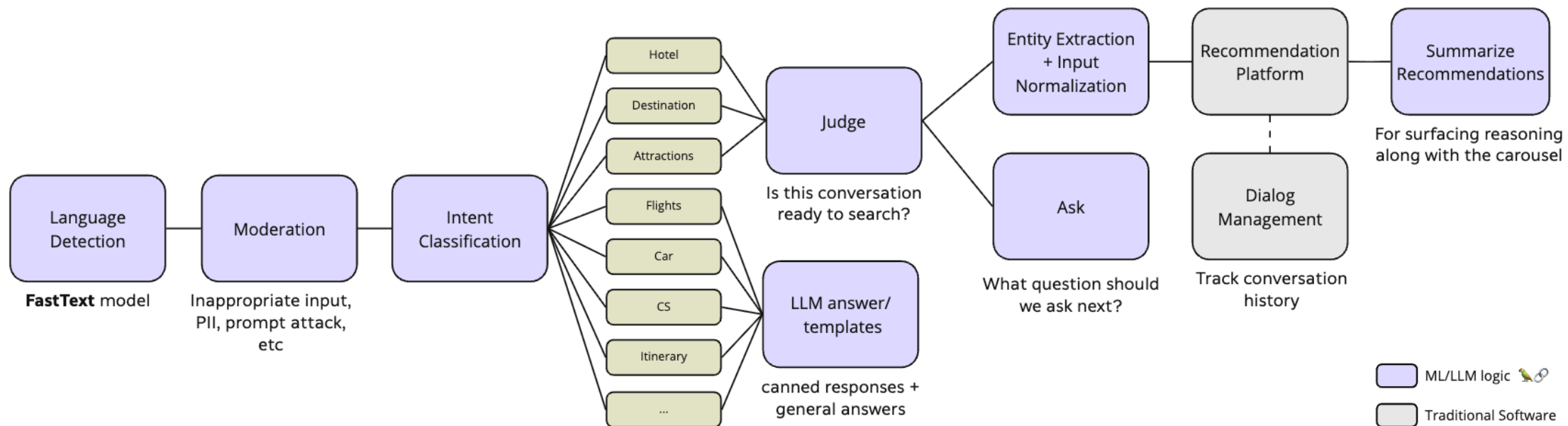
Hallucinations
were a daily reality

Tool calling
was unavailable

Structured outputs
were unreliable

Agent patterns
did not exist yet

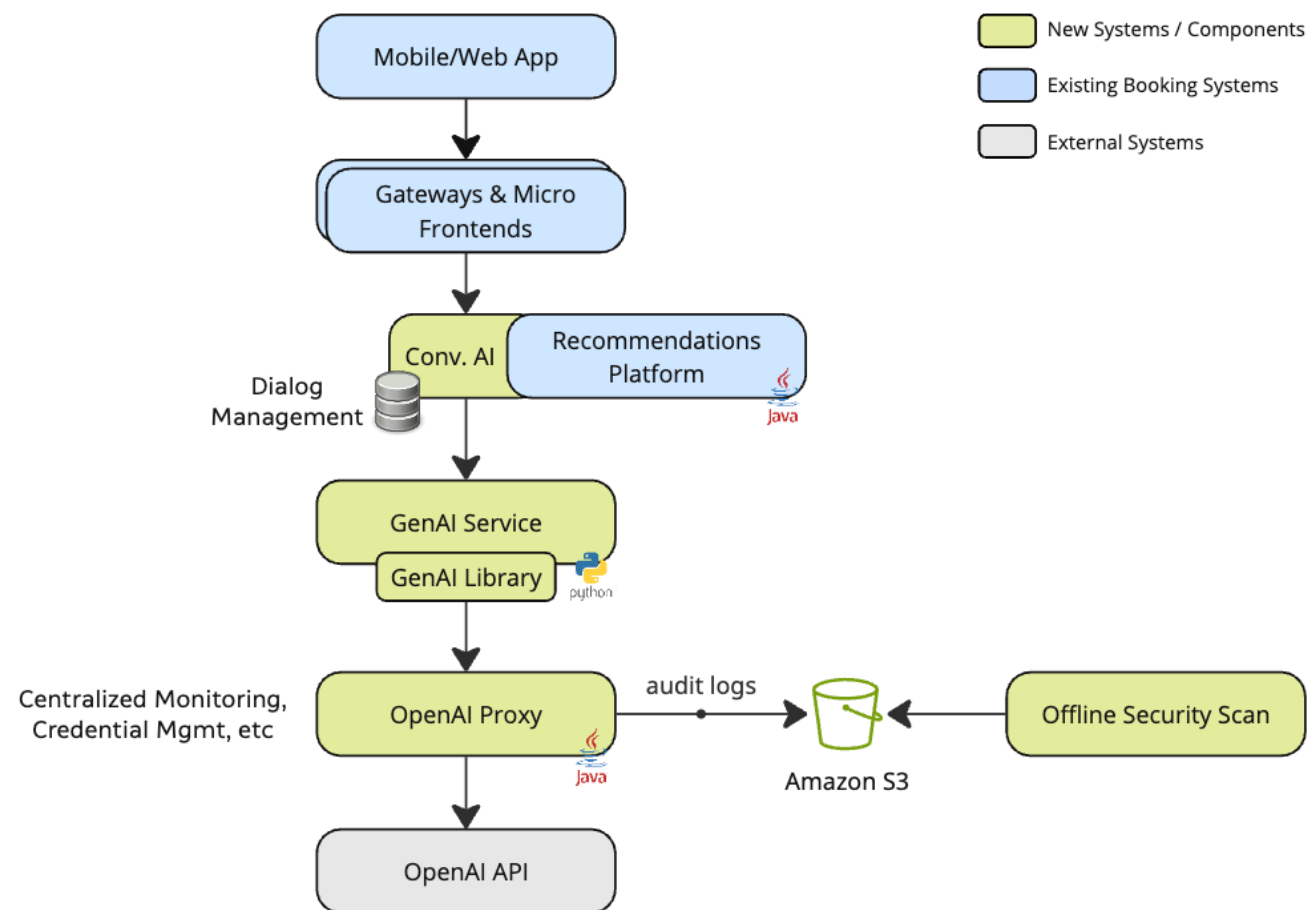
AI Trip Planner's flow



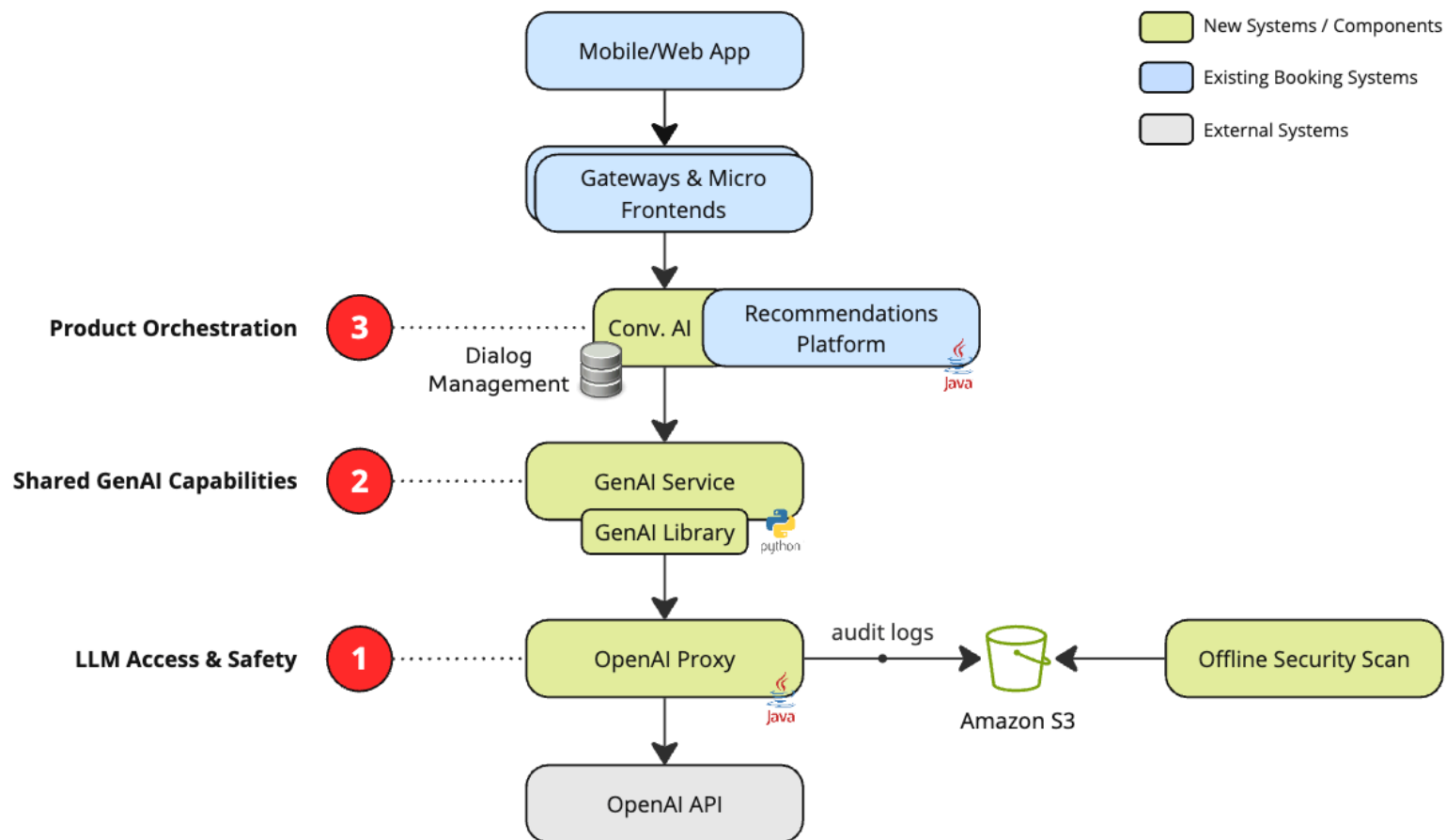
THE OPERATING MODEL

A temporary **tiger team**
8
weeks

The intial architecture

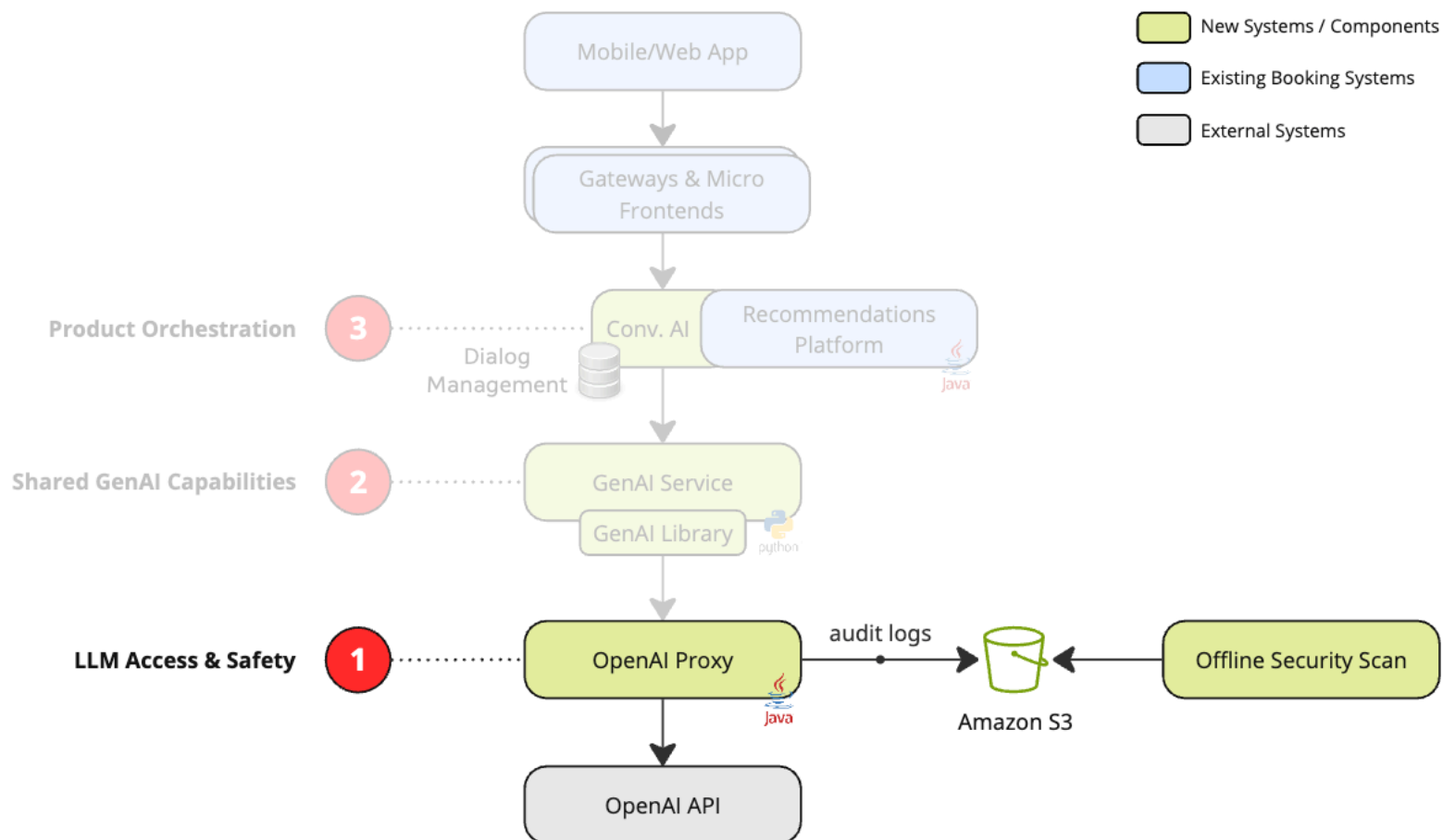


Three layers of evolution



Layer 1 evolution

LLM access and safety



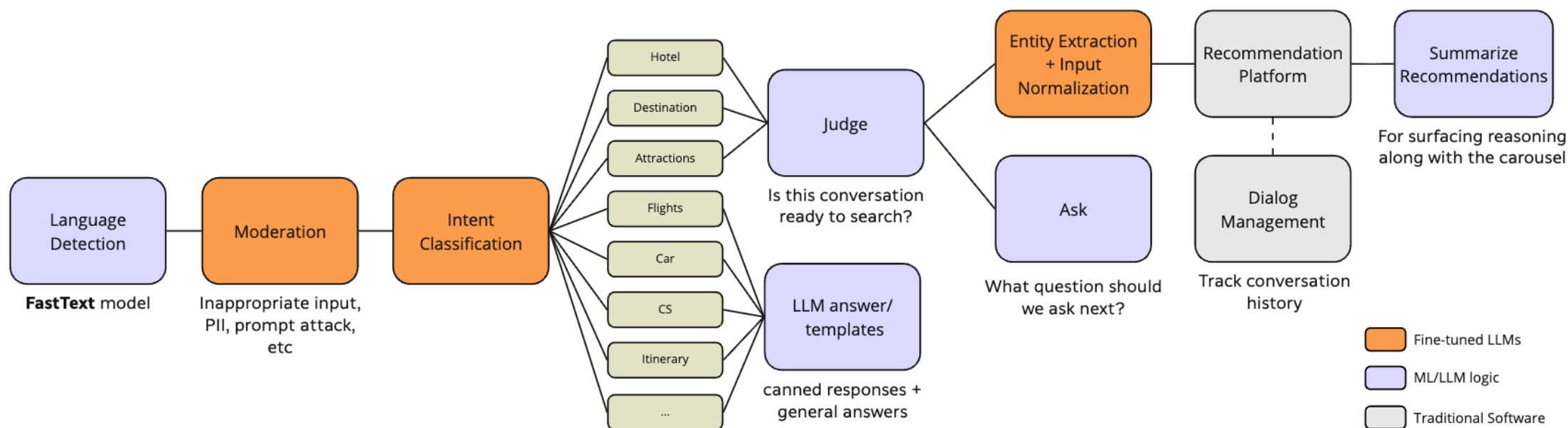
THE NEXT CHALLENGE

Model accuracy became a priority.

As the product matured, accuracy became an important concern.

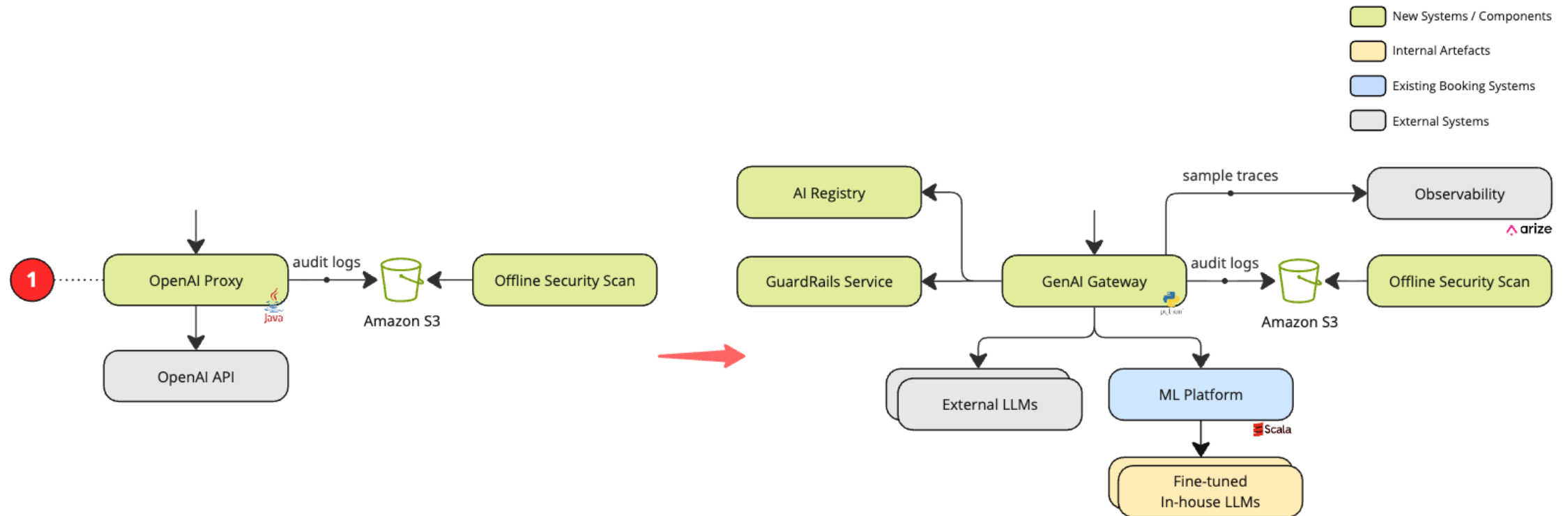
We started fine-tuning models

To improve accuracy, latency and cost efficiency



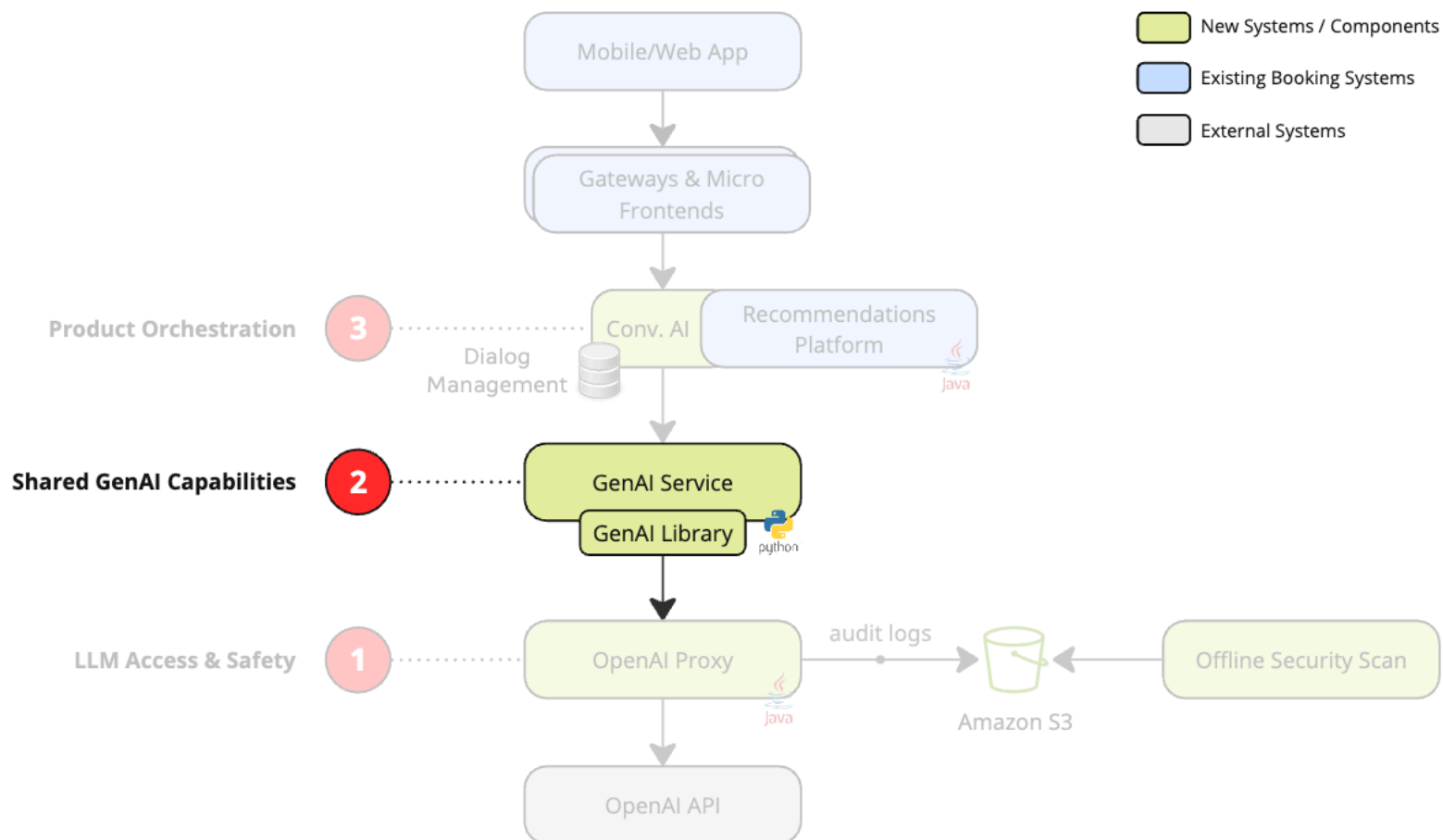
GenAI Gateway

An unified interface for 3rd-party and in-house models



Layer 2 evolution

Shared GenAI capabilities

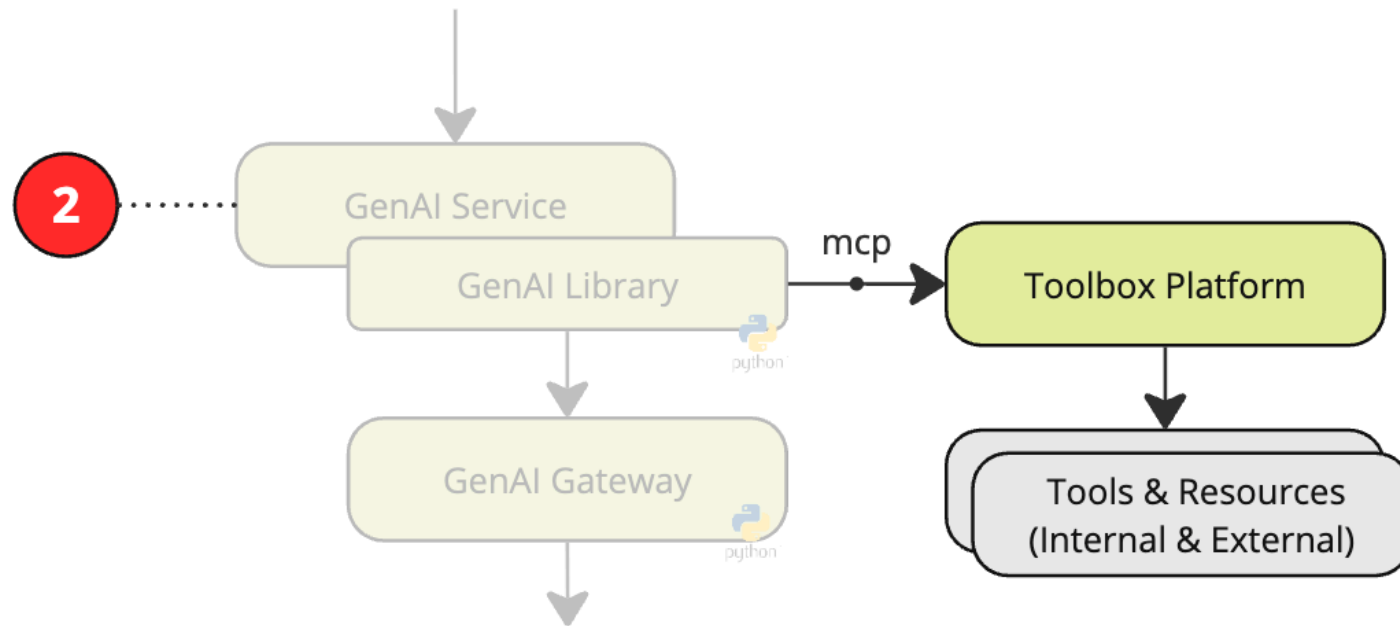


THE NEXT ABSTRACTION

We started building **agents**

Connecting Agents to Tools

A governed access layer



DEMOCRATIZING AI

Enable everyone to **build with AI**

DEMOCRATIZING AI

Two routes to build AI agents

Reusable agents

Configure and deploy quickly.

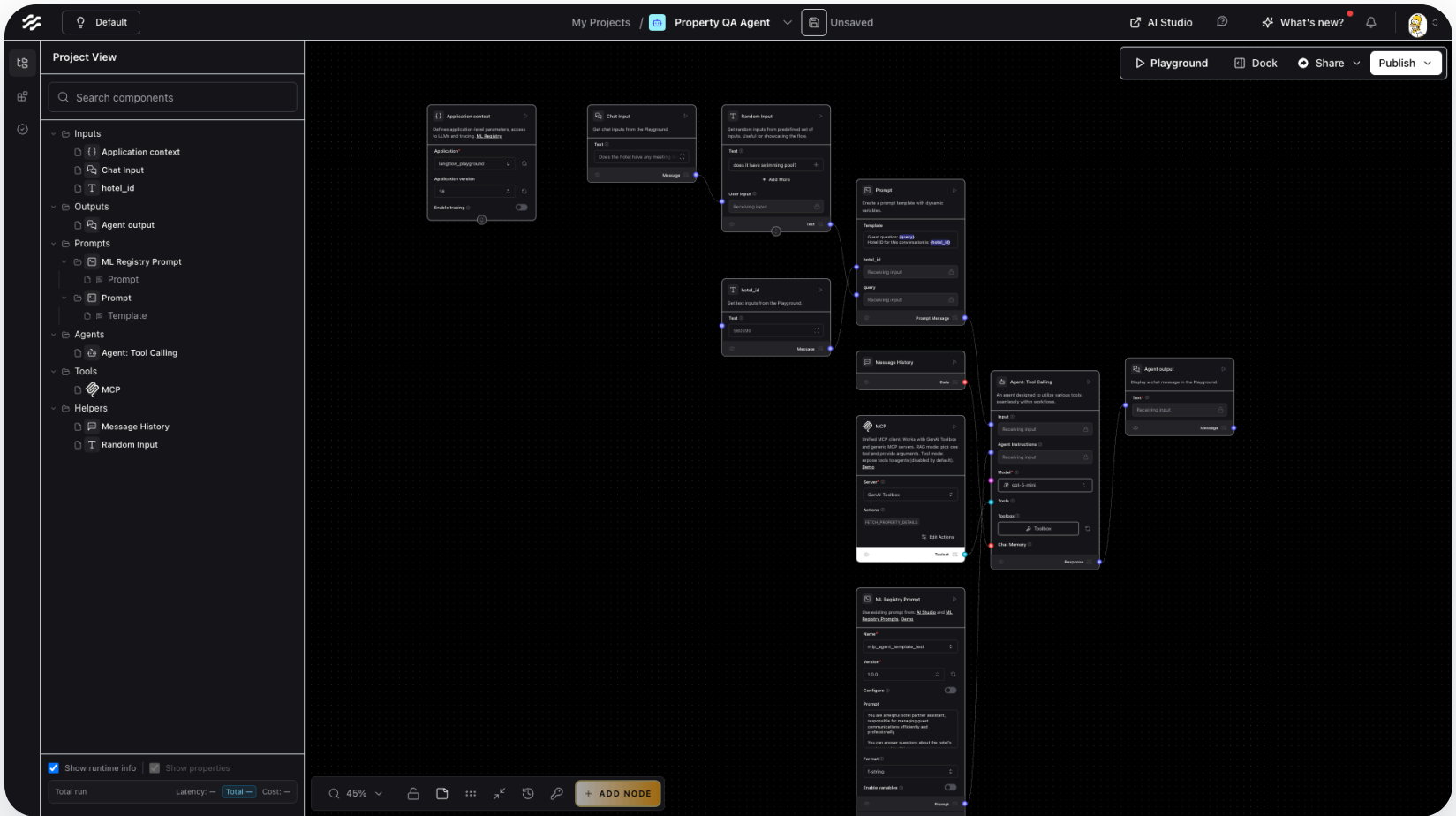
Deep customization

Build and deploy end to end.



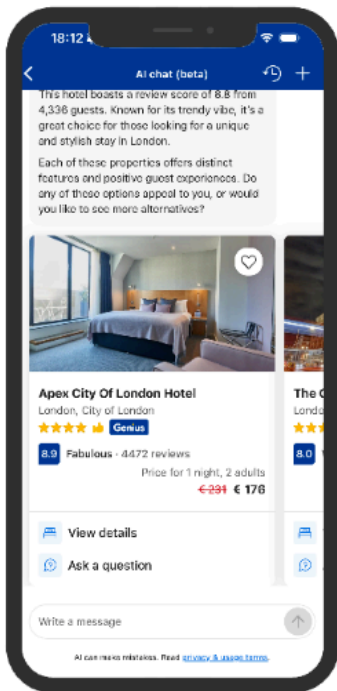
AI Studio

Prototype, test and deploy agents in a visual manner

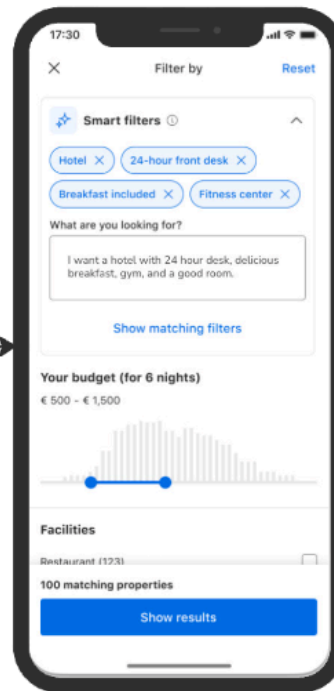
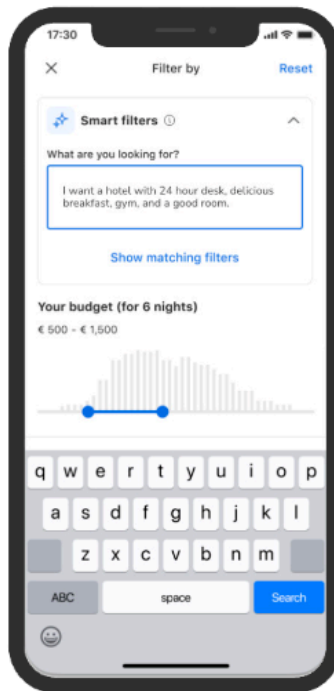


GenAI use cases

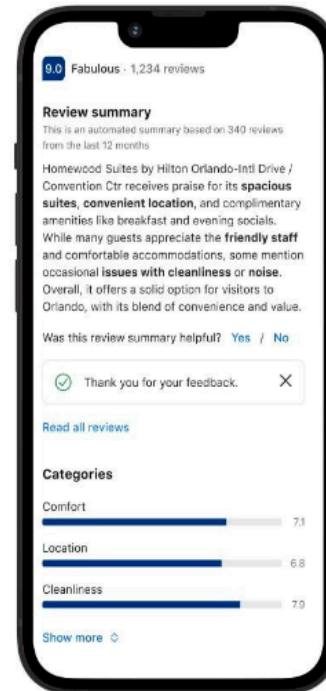
Trip Planning



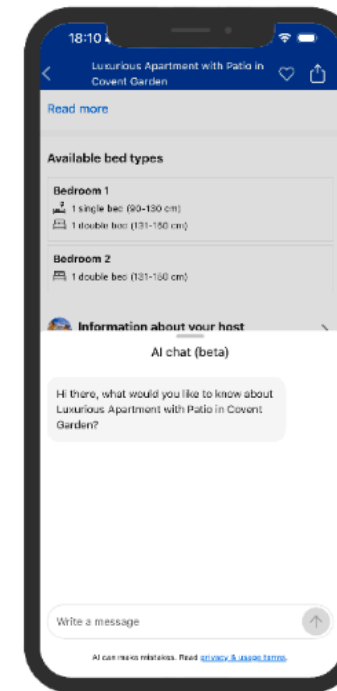
Smart Filters



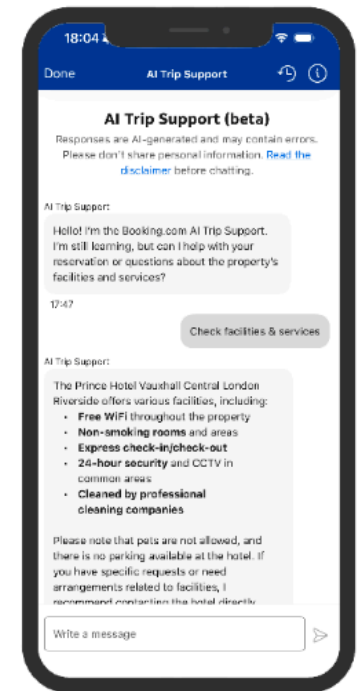
Review Summaries



Property QnA



Post-Booking Support



THE PLATFORM NEED

**We wanted deep customization
without redundant effort.**

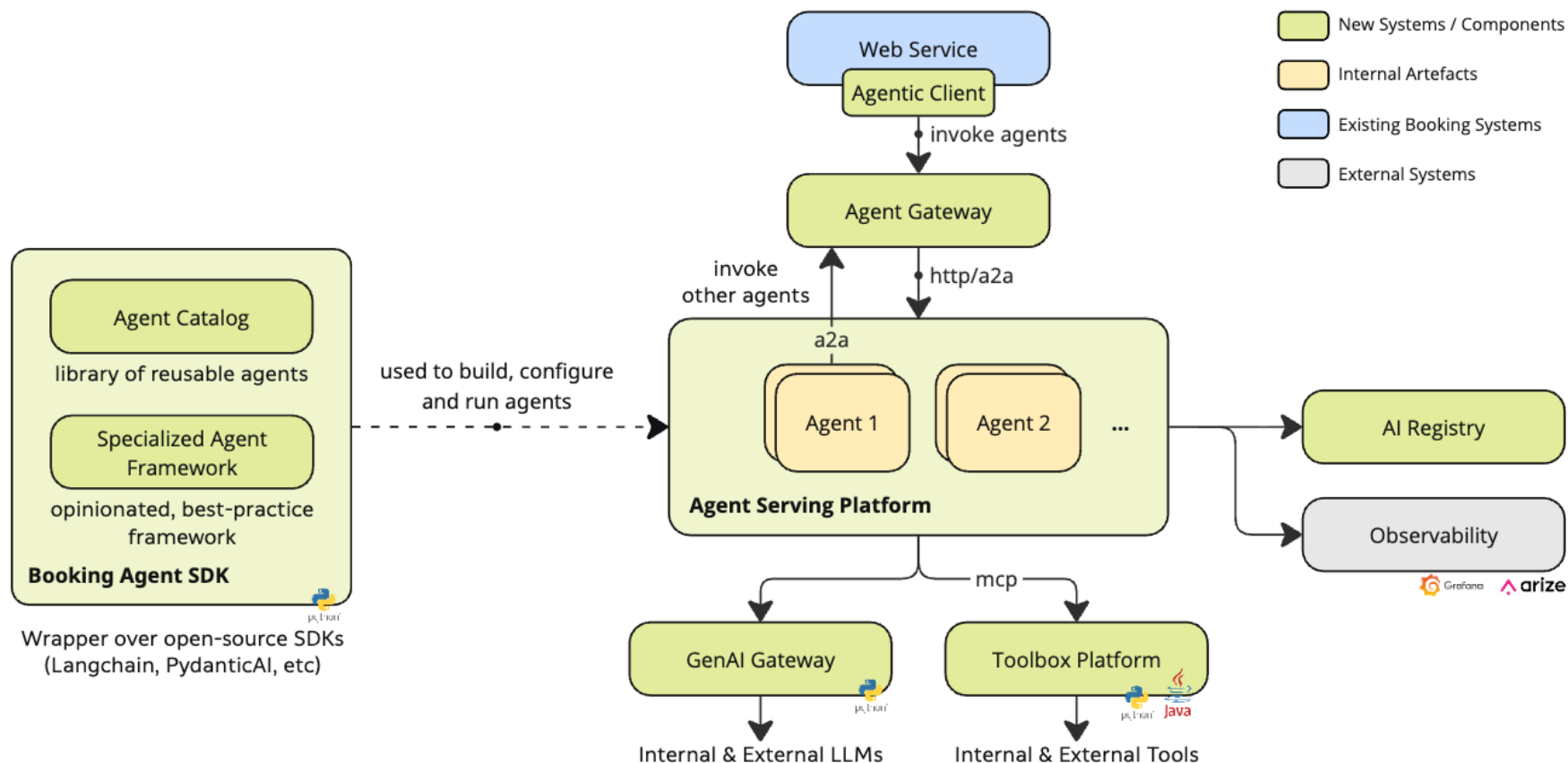
MANAGED AGENT RUNTIME

Agent Serving Platform

Run **custom Python agents**

Agent Serving Platform

A platform to build and run agents



Other foundational systems

Supporting services behind reliable agent execution

Short-term Agentic Memory

Enables LLMs maintain ephemeral context across sessions

Embedding Service

Provides multi-model multi-provider abstraction

RAG Service

Config-driven RAG as an API

THE NEXT PLATFORM CONCERN

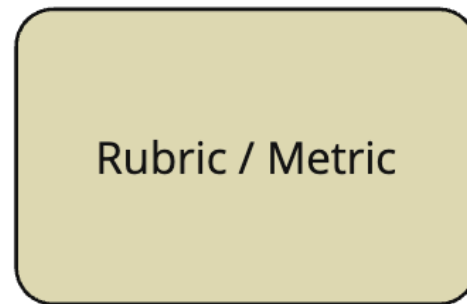
Evaluations

Ensuring reliability in a probabilistic world

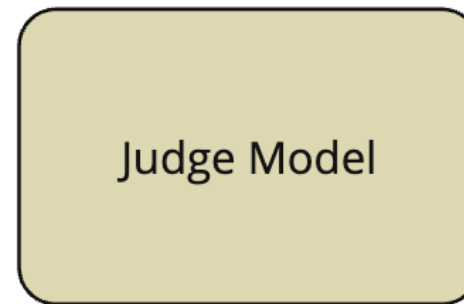
LLM AS A JUDGE

Using **an LLM to judge** the output of another LLM

The building blocks of evaluation

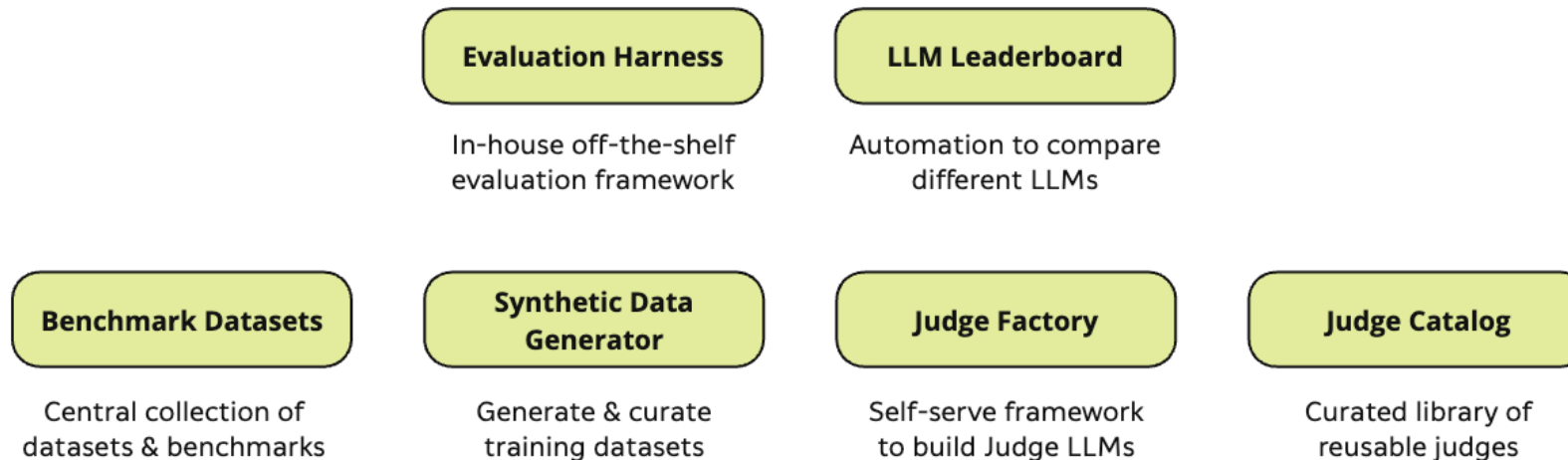


The evaluation criteria



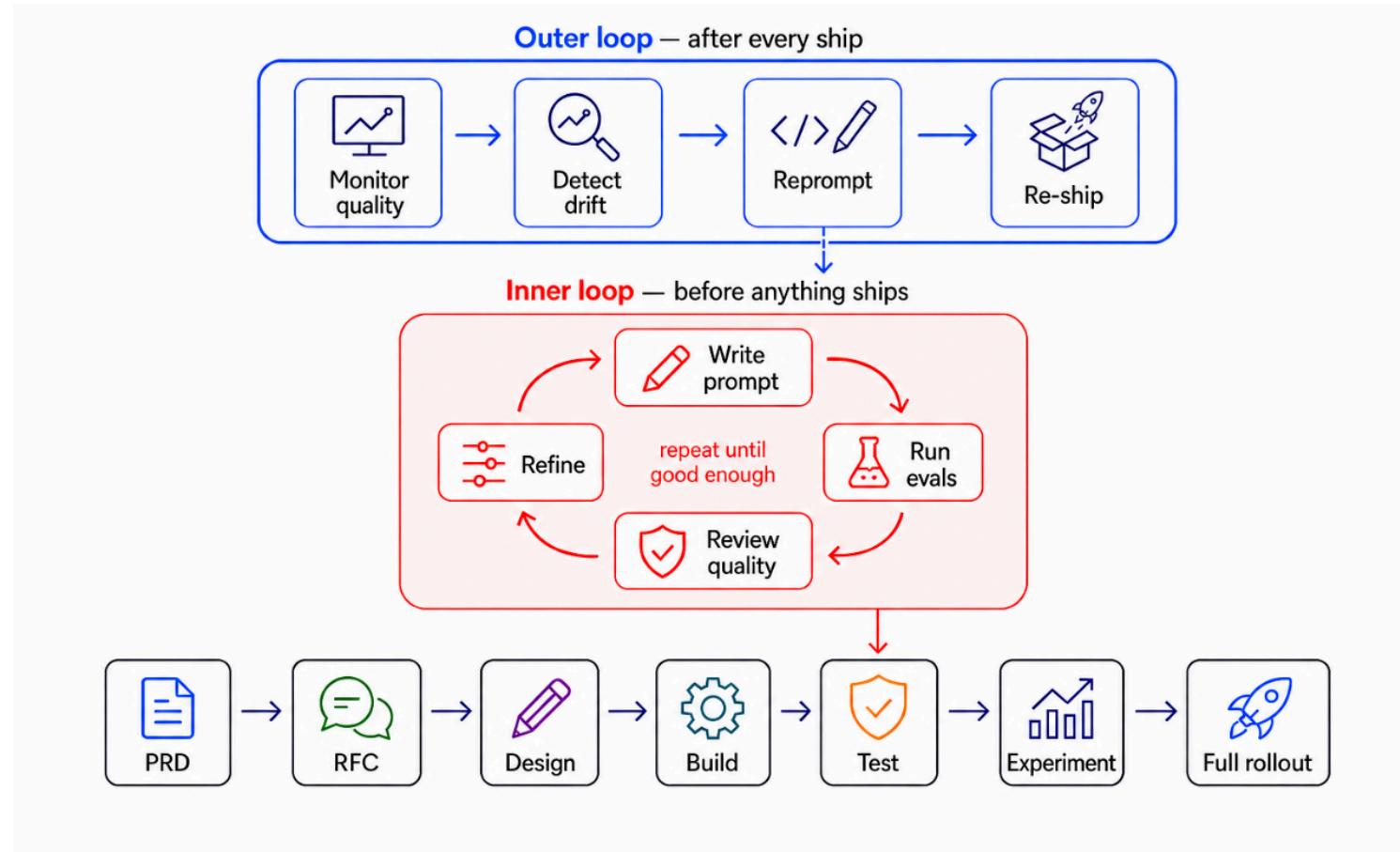
The LLM that applies it

The evaluation stack

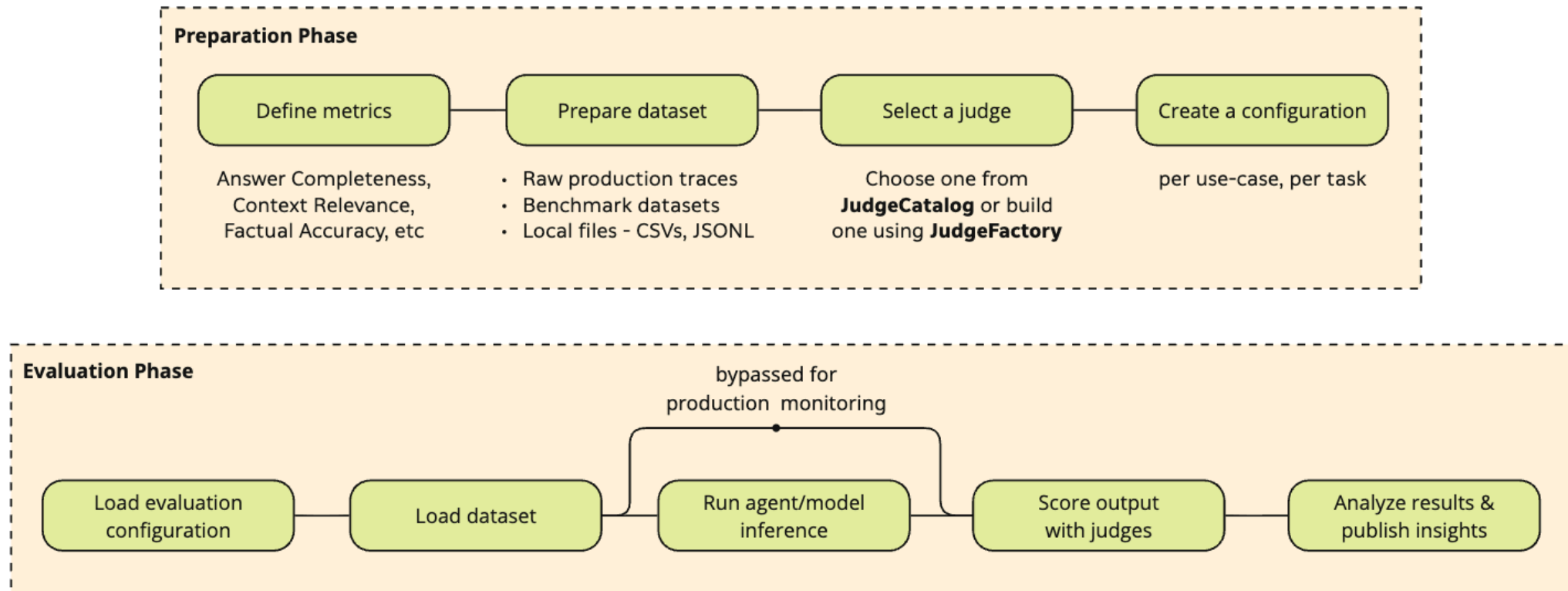


Offline and online evaluation

Benchmark before launch and monitor after release

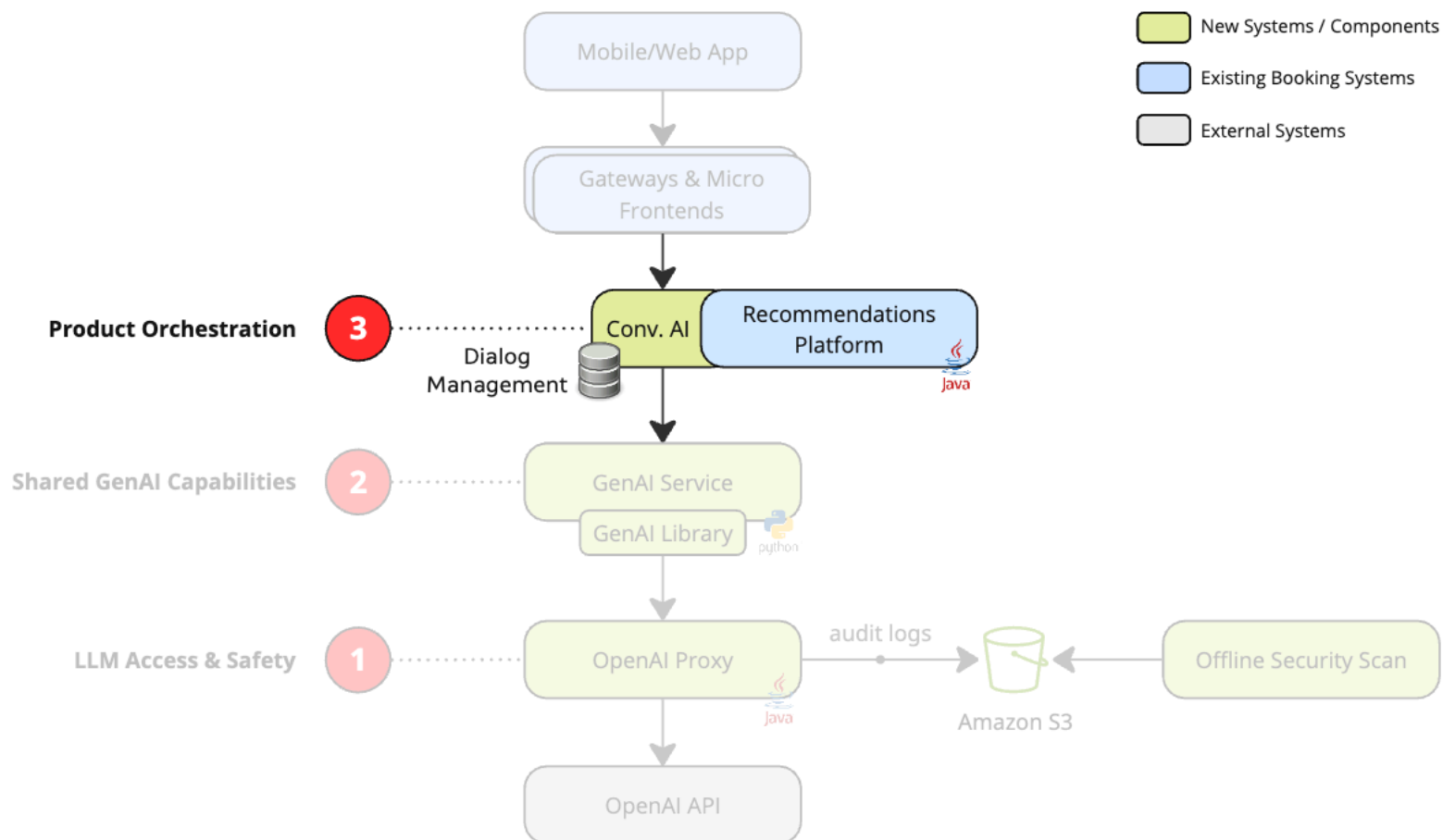


Automating evaluations at scale



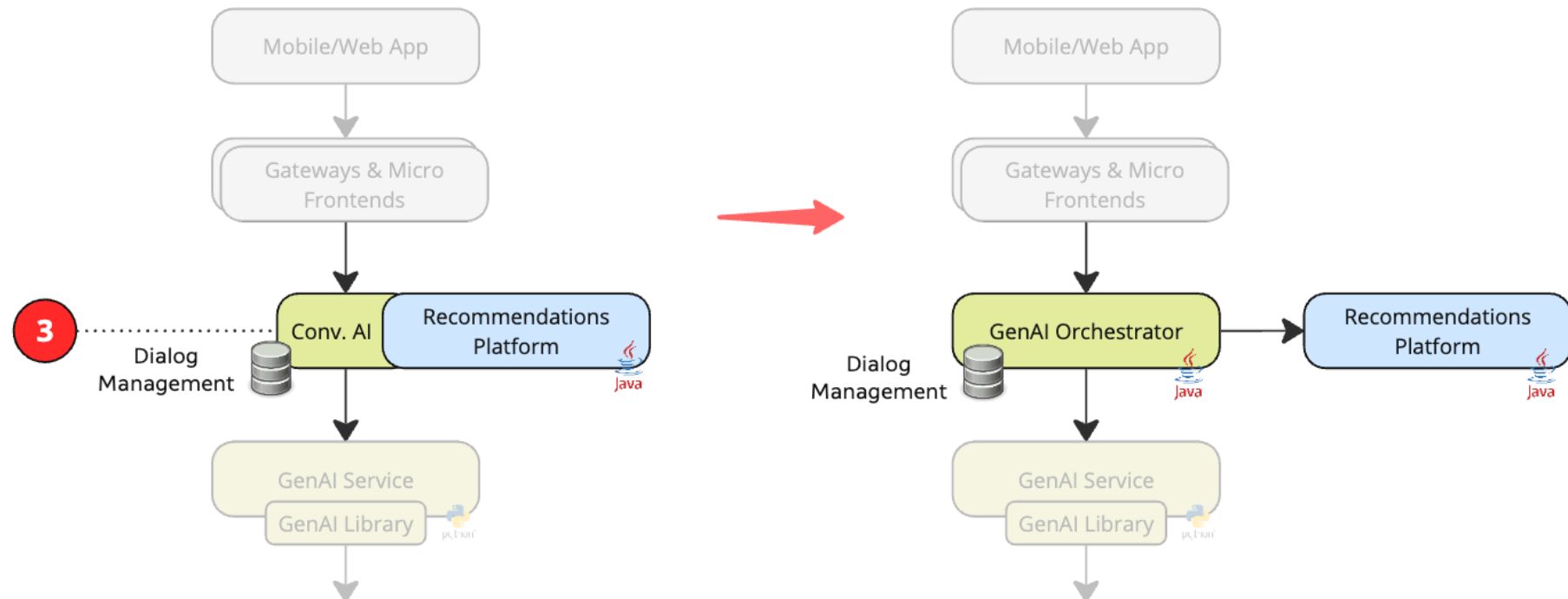
Layer 3 evolution

Product orchestration



A dedicated GenAI Orchestrator

Product logic moved into a specialized workflows



Orchestration as configuration

A custom DSL to define GenAI workflows as directed graphs

```
JSON
{
  "schema": "workflow/v0",
  "name": "property_qa",
  "description": "Answers questions about a property",
  "type": "agentic",
  "graph": {
    "nodes": [...],
    "edges": [...]
  },
  "input_output_info": {
    "input_schema": "property_qa_input",
    "output_schema": "property_qa_output"
  },
  "state_providers": []
}
```

ORCHESTRATION AS CONFIGURATION

It worked, but it was **not perfect**

- Weak typing
- Unsafe I/O mappings
- Execution engine constraints
- Limited debuggability and observability

THE NEXT ITERATION

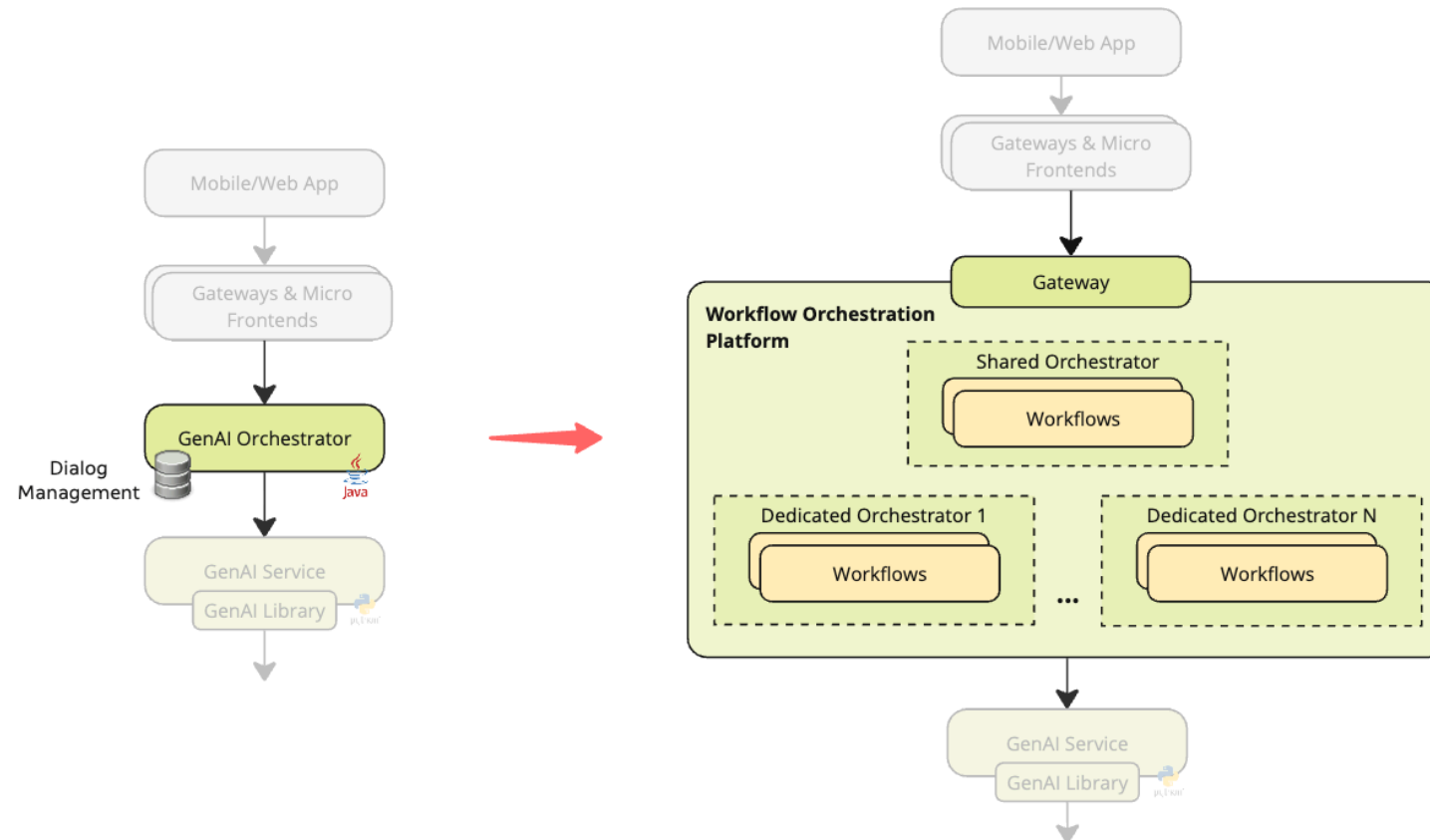
So, we optimized it further with
a **code-first** approach

```
graph LR; subgraph SDKs [Orchestration SDK]; direction TB; JS[Java SDK]; PS[Python SDK]; end; JS -- generates --> IR[Intermediate Representation .proto]; PS -- generates --> IR; IR -- used for --> subgraph Ops [ ]; WE[Workflow Execution]; WR[Workflow Registration]; WV[Workflow Validation]; WVis[Workflow Visualization]; OASG[OpenAPI Spec Generation]; Ellipsis[...]; end; GT[Graph Transpiler] -- read --> JG[JSON graph Legacy workflows];
```

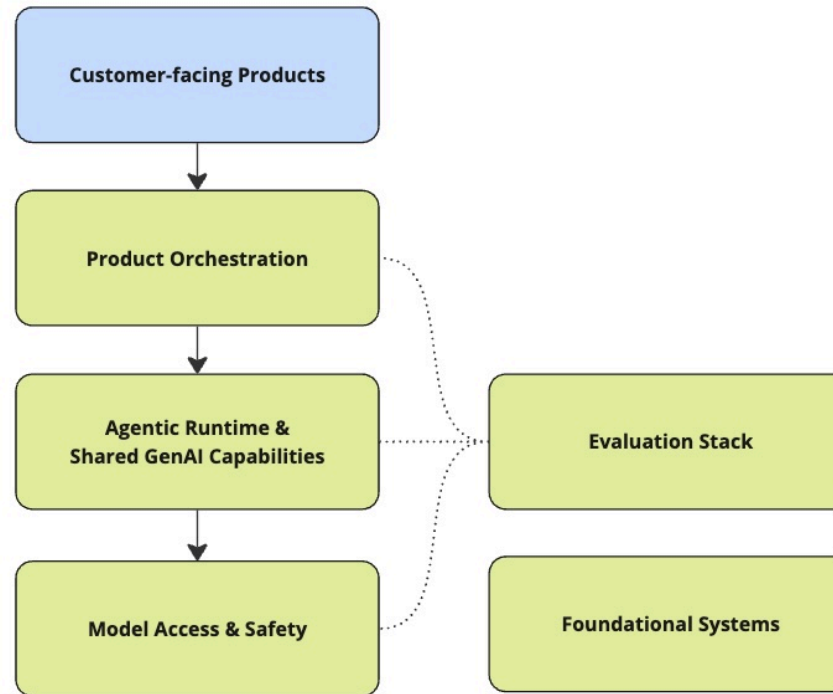
The diagram illustrates the architecture of the workflow engine. It starts with the **Orchestration SDK**, which contains the **Java SDK** and the **Python SDK**. Both SDKs **generate** an **Intermediate Representation .proto**. This representation is then **used for** various operations: **Workflow Execution**, **Workflow Registration**, **Workflow Validation**, **Workflow Visualization**, **OpenAPI Spec Generation**, and others. Additionally, the **Graph Transpiler** **reads** **JSON graph (Legacy workflows)** and converts them into the intermediate representation.

Workflow Orchestration Platform

Moving towards self-service



Recap: A conceptual view



THE TAKEAWAYS

What we learned

Three lessons from building the ecosystem.

WHAT WE LEARNED

Start with the product, not the platform

Build first. Learn. Generalize what repeats.

WHAT WE LEARNED

Limit autonomy to what you can validate

Reliable AI needs bounded workflows

WHAT WE LEARNED

Evaluate before and after launch

Build evaluation into the product loop.



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Principal Engineer @ Booking.com



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